

# A Conceptual Model to Support Teaching of Software Engineering Controlled (Quasi-)Experiments

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## ABSTRACT

Throughout controlled experimentation, it is possible to provide evidence of the software being developed. In the academic environment, Experimentation in Software Engineering (ESE) is essential to understanding cause-effect relations, enabling a vision of the development process, and taking action on actual events in the software industry. As much as the experimentation processes have been used in industry and academia, there is a lack of formalization of the principles of ESE teaching and artifacts that can be useful to support it in higher education. One of the means to contribute to such a topic would be the design of a conceptual model, which is widely discussed in the literature, thus applying empirical methods for a better understanding of the context and representation of ESE teaching. Thus, in this paper, we developed a conceptual model to support the teaching of controlled experiments and quasi-experiments. To design the conceptual model, we carried out an analysis of metadata from controlled experiments and quasi-experiments in the literature and conducted a survey to collect data from instructors who teach ESE. We evaluated the model with the Technology Acceptance Model (TAM). Results consist of a feasible conceptual model aiming to standardize the basic concepts of ESE and further support the production and reuse of ESE materials.

## CCS CONCEPTS

• **General and reference** → Experimentation; Empirical studies; • **Software and its engineering**;

## KEYWORDS

Teaching of Controlled Experimentation, Controlled Experimentation, TAM Model, Conceptual Modeling, Concepts

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## 1 INTRODUCTION

Experimentation is the basis of the scientific method, a systematic way of exploring the world [8]. Although scientific experiments are usually associated with laboratories, they can take place anywhere, at any time, including in the software development context [15].

In Computer Science, Experimentation in Software Engineering (ESE) plays a central role in the communities of Software Engineering (SE) researchers and professionals. The ESE is intended to promote evidence of a certain theory or technology related to the software development life cycle [29].

It is essential to highlight how the knowledge gained from a controlled experiment or quasi-experiment<sup>1</sup> can be important in the broader development context when involving students. Controlled experiments can also be important from a research or technology transfer perspective. Anyway, there is a need to improve the way we teach and conduct experiments in software engineering [27].

Experimentation should be an indispensable part of higher education research. As Tichy [24] says, “Experimentation can help to build a reliable knowledge base and thus reduce uncertainty about which theories, methods and tools are suitable”.

Basili et al. [3] state that an empirical study, in a broad sense, is an act or operation with the purpose of discovering something unknown or of testing a hypothesis, and may encompass diverse research strategies, including all forms of experiments, qualitative studies, research, and analysis. An experiment is a form of empirical study where the researcher has control over some of the conditions under which the study takes place and control over the independent variables under study.

According to Kuhrmann [17], ESE can be applied to: (i) analyze/understand phenomena in software development, (ii) identify/assess the strengths and weaknesses of software engineering approaches, and (iii) investigate the state-of-the-art/practice to identify promising solutions/approaches.

The higher education teaching process currently requires specific points of attention to ensure quality in teaching and learning for students, bringing them closer to the real world, and providing to go beyond theory. Instructors are one of the main ones responsible for teaching, researching, and managing the educational process. They choose the methods used to capture the attention of their audience in a given period. The instructor must use didactic instruments prioritizing student participation, and managing their expectations and abilities [5].

In this context, the software engineering community has discussed how to transfer practices and experiences to students. The

<sup>1</sup>From now on, we refer “experiment” to both “controlled experiment” and “controlled quasi-experiment”.

root of the problem seems to be how SE is usually taught. Drawing on theories and concepts presented in a series of lectures or classes, students are required to develop a small fictional project in an attempt to put this newly discovered knowledge into practice [2].

Hilburn and Towhidnejad [11] state that in many cases software companies have to complement the knowledge of recent graduates with training and provide skills and technical competencies with a focus on the development of software systems.

It is believed that ESE can help the software industry demand more qualified human capital with experience. In the literature, there are materials that report experiments, but in a disorderly way or not following a specific guideline, nor do they present a formalization of necessary basic principles about teaching experiments in SE [9, 20, 26].

Faced with these problems, the motivation arises in developing a model to support the teaching of ESE, seeking to present the necessary aspects for the teaching of experimentation. A conceptual model is a representation (usually graphical) constructed by someone with domain perception. It can be used to facilitate the design and implementation of an information system. Such a conceptual model is then validated by stakeholders against the represented concepts [23]. Therefore, we provide a conceptual model developed to support the teaching of controlled SE experiments and quasi-experiments. Such a model has the function of providing standardization of concepts about SE experiments, as well as supporting the production and reuse of teaching materials on experimentation.

The conceptual model was elaborated with the view of contributing specifically to the formalization and structuring of data and metadata about ESE teaching. Thus, it can serve as a basis for other applications in future work, such as the construction of an ontology and/or a framework that can make inferences in materials for teaching ESE.

## 2 BACKGROUND AND RELATED WORK

This section presents essential concepts to understand this work and discusses related work.

### 2.1 Experimentation in Software Engineering

The use of experimentation aims to seek facts through assumptions or speculations regarding a given subject. In the software engineering area, experiments are usually related to software development or maintenance. An experiment is performed to answer assumptions or speculations, demonstrating the real importance of this method of study/research [30].

The main reasons for carrying out quantitative studies are the opportunity to obtain objective and statistically significant results in relation to the understanding, control, prediction, and improvement of software development. Empirical studies are an important input for decision making and have a better organization of the collected data [29].

The conduction of an experiment is based on the relationship between cause and effect, through a theoretical model between two or more related phenomena to determine whether the proposed model can be considered correct. Thus, the model is used to create a hypothesis regarding the particular changes in the phenomena (the

cause) that will lead to changes in the other (the effect). Therefore, the role of the experiment is to test the hypothesis, thus refuting the null hypothesis or not [16].

When conducting a controlled experiment, we want to study the result by varying input data. Thus, there are two main types of input variables, namely [16]:

- Independent Variables, which are associated with the cause and controlled as a result of the experimenter's activities. When the experiment studies the effect of altering one or more independent variables, they are called factors that can assume values called treatments; and
- Dependent Variables, which are associated with the effect and result from the changes that the experimenter makes to the independent variables.

In the area of Software Engineering, especially, it is difficult to perform true experiments, that is, those that have complete randomness, because of the difficulty of allocating participants and/or objects to different treatments in a random way. Thus, experiments in this area are often quasi-experiments, in which there is no randomness of the participants and/or the experimental objects, for instance, software artifacts as processes, tools, or source-code [30].

Regardless of conducting experiments or quasi-experiments, both should follow the same processes and steps as they are important means to be used as checklists and guidelines. Furtado et al. [9], for instance, provide guidelines to support proper documentation of SE experiments, which is a direct result of works on improving SE experimentation [20, 26].

### 2.2 Teaching Experimentation in Software Engineering

Software-based solutions are part of people's daily lives, making them dependent on their uses in today's society. With the due importance of software in today's world, there is a concern with the quality of the software process, causing more attention to be given to the way students learn their concepts and processes [28].

In the existing literature, there are several discussions on problems in the teaching-learning process in the SE area, thus the need for greater integration and cooperation can be clearly observed. Such discussed problems require greater integration and cooperation among instructors. In SE teaching, it is a major challenge to integrate applied methodology and theory into software development practice, rather than learning concepts and methodologies as abstract ideas, as discussed by Dahiya [5].

Host [12] states that, as in many other fields of applied research, there has been a connection between education and empirical methods in software engineering. In many cases, software engineering students and computer science students are involved in empirical studies.

In certain contexts, higher education is based only on theoretical concepts, not applying meaningful practice, which can be harmful to students after graduation. Gnatz et al. [10] claim that the only way that really prepares someone to do a successful work project is to learn by doing. From the university's point of view, it is a challenging task to find a teaching method to provide students with practical knowledge about the labor market. In addition, Santos et al. [22] say that the students present in the courses in the area of

Computing point out the challenge of innovation in recognizing and accommodating the competencies and abilities of students in higher education, because of its breadth of research and application.

According to Host [12], “Decisions must be made based on some type of investigation of the suitability of the new technique in the specific organization”. Students often lack the opportunity to experience real problems (e.g., sizable systems or large-scale projects with clients) or to work in an environment that supports learning from failure (e.g., working under severe stress, handling issues in teams or project failure) [18].

Cagiltay [4] reports that industry and university environments have changed. Industry demands skills belonging to professionals which also changed. The practice in industry differs from what was previously required and teaching at the university. For this reason, students must be able to apply concepts learned during their education to real-world problem-solving. Therefore, ESE is seen as a learning practice from an evidence-based pedagogical perspective. Since many practitioners face problems of changing techniques or methods in software development organizations, it is necessary to teach students how empirical studies are carried out, guiding them in a direction to generate reliable evidence and inferences for decision-making [22].

Experimentation offers a systematic, disciplined, computable, and controlled way to evaluate human activity. New methods, techniques, languages, and tools should not be suggested, published, or presented for sale without experimentation and validation [7, 14].

According to Wohlin [28], there are different ways of teaching ESE in a given curriculum, which are:

- integrated into higher education courses, in which ESE is part of other subjects and is evaluated based on learning and assignment assessments; and
- as an isolated discipline, in which the biggest advantage lies in the specific focus on experiments as practical work.

According to Wohlin [28] a particular challenge is balancing the different goals when conducting empirical studies in academic curricula. If a study is part of a course, then research should not dominate educational objectives. In another case, it is not just a question of research; it is also a matter of using an empirical study in an academic context as a springboard in a technology transfer process.

Involving students in experimentation is not an easy situation. Teaching through quality teaching materials leads to better learning. However, listing which models of experiments are more suitable becomes an unfeasible task, without any support system for pedagogical resources.

The proper teaching of the concepts of experimentation in software engineering will make the software industry a human capital with knowledge about experiments, however, no didactic material or formalization of ESE teaching is found, thus providing an opportunity to explore this subject.

### 2.3 Related Work

This section discusses related work regarding teaching experimentation in SE.

The work of Travassos et al. [25] reports a computerized infrastructure to support large-scale experimentation and the management of scientific knowledge in software engineering. The authors present issues in conducting studies due to the lack of models, formalization, or conceptualization of ESE. Such work evidences the need for the construction of means to support the teaching of ESE, as well as in the formalization of concepts and directions of resources.

Alves et al. [1] developed a process structure, involving students as subjects of empirical studies of software engineering. The work contributes to understanding the context of the need for materials to support ESE teaching, such as the development of educational resources that could aid instructors in the ESE teaching-learning process.

In the work by Kuhrmann and Münch [18], the reflection on SE courses is seen in the context of providing knowledge and skills that are related to market practice. Such work is analyzed from the point of view of the importance of the student’s experience in practices that can collaborate with their training process at the university, presenting the importance of ESE. However, such work does not approach aspects of ESE teaching or indication of materials necessary in the academic learning process.

By analyzing these related studies, we can understand there is an opportunity to investigate and specify a conceptual model to support the teaching of ESE as a preliminary artifact to aid instructors due to the number of concepts involved.

## 3 THE CONCEPTUAL MODEL FOR ESE TEACHING

In this section, we present our conceptual model to support the teaching of ESE.

We carefully designed our conceptual model (CM) (see section “Artifact Availability”) based on existing literature studies on ESE conceptualization [9, 19] and a review on Open Educational Resources (OER) and how to build open scientific knowledge. We seek to identify practices and real instruments (materials) used to teach ESE to undergraduate and graduate students.

Our CM might be used: (i) as a checklist of the needed elements to thoroughly better plan, conduct, analyze, and document experiments; (ii) to increase and promote reproducibility by providing well-documented experiments and decision-making; (iii) as a template to be used by instructors in creating and evaluating new ESE courses with a common ground and vocabulary.

The developed CM is represented by elements that can be used or explored by instructors for the teaching of controlled experiments and quasi-experiments. Such elements serve as a basis for which resources can be used, being represented throughout concepts and relationships. Some concepts are trivial and easy to understand, while others are generally implicit and require a greater analysis of the context.

We took into account the separation of concerns for modeling the CM into artifacts that can be used by the instructors. Each artifact is related to one or several concepts that are considered important in the domain. Thus, the modeled concepts are grouped in the CM and graphically represented by ellipses and respective relationships.

A relationship represents an association among elements. In the CM, associations are seen graphically as the connecting lines among groups and concepts. Certain associations are quite simple to identify, and their roles (see Figure 1) are demarcated with a verb that can be applied to certain basic rules in the ESE teaching process. Other associations are consequences or complementary actions of the groups of implicit concepts and may require more experience from the instructor.

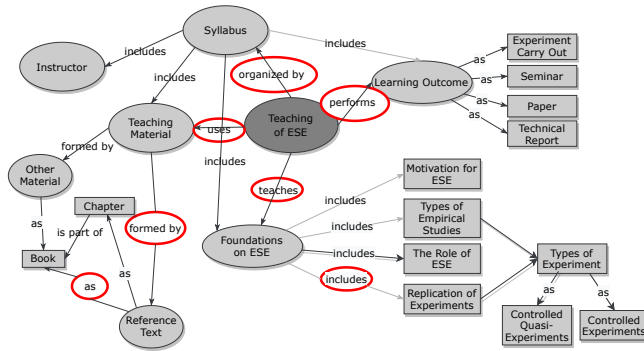


Figure 1: Concepts Relationships and Roles

We designed the relationships in the CM to give meaning to the actions by going through the points of connections and interests until reaching the concepts that are grouped, thus being able to instantiate our CM for a given ESE course.

Figure 2 depicts the CM main concepts. The relationship among elements is proposed as a way to guide instructors on the resources and concepts related to ESE teaching. By using the CM, instructors can understand how each group is related to other groups of concepts.

In Figure 2, every ESE course (**Teaching of ESE**) must provide a **Syllabus**, which indicates: one or more responsible instructors (**Instructor**); the teaching materials (**Teaching Material**), which might be composed of **Reference Text** and/or **Other Material**; the content to be taught (**Foundations on ESE**), and how learning will be verified (**Learning Outcome**). **Other Material** can be also supported by **Open Educational Resource (OER)**.

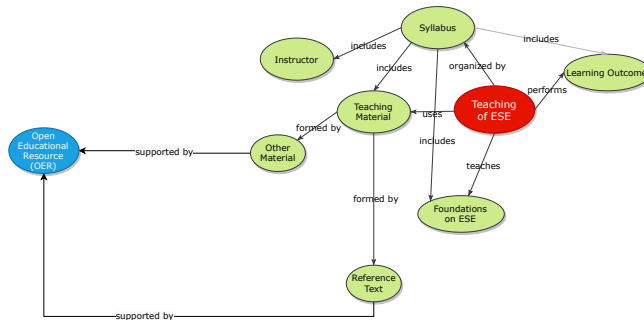


Figure 2: The Conceptual Model Main Concepts

The grouping of concepts and the designed relationships aimed to make the CM easy to understand for both newcomers and experienced instructors on teaching experimentation, thus being able to interpret and instantiate the model for the concepts to be taught.

The remaining concepts will be illustrated in the next sections.

### 3.1 Syllabus Concept

As a method of organizing the Teaching of ESE, the concepts to be taught are organized in the form of a Syllabus. A syllabus is a student guide to a course and provides information on required texts, assignments, and instructors.

This concept includes Instructor, Foundations on ESE, Learning Outcome, and Teaching Material, as we can see in Figure 2.

### 3.2 Foundations on ESE Concept

Figure 3 depicts the group of concepts for Foundations on ESE, composed of:

- **Motivation for ESE:** aims at motivating students on the importance of experimentation for promoting reliable evidence in their research;
- **Types of Empirical Studies:** refers to providing students essential concepts on the types of empirical studies, such as surveys, systematic and mapping reviews;
- **Types of Experiment:** refers to provide essential and further concepts on both controlled experiments and quasi-experiments;
- **The Role of ESE:** should introduce students to the central role ESE should play as the scientific method for the software industry and academia;
- **Replication of Experiments:** aims at teaching the basics of replicating experiments following all the steps reported in the respective literature to enforce previous, original evidence;
- **Mathematical Foundations:** should encompass essential contents to make students able at comprehending and applying basic quantitative analysis as, for instance, theory of set and data tabulation;
- **Basics of Probability and Statistics:** content referring to the basic aspects of probability and its consequence for statistical techniques and analysis as, for instance, descriptive and inferential statistics;
- **Simulation Research:** encompasses techniques and tools to conduct experiment simulations.

Types of Experiment is the specification of the concepts of Controlled Experiments and Controlled Quasi-Experiments.

Such concepts are directly related to the concept Controlled Experiment/Quasi-Experiment. Such group was based on Furtado et al. [9] and has as main concepts (Figure 4) Planning, Operation, Experiment's Dissemination and Analysis and Interpretation. These concepts are related to the proper documentation of controlled experiments.

### 3.3 Learning Outcome Concept

One of the processes of teaching experimentation involves the analysis of students' academic performance on the concepts presented. This is possible with documents and tasks that represent

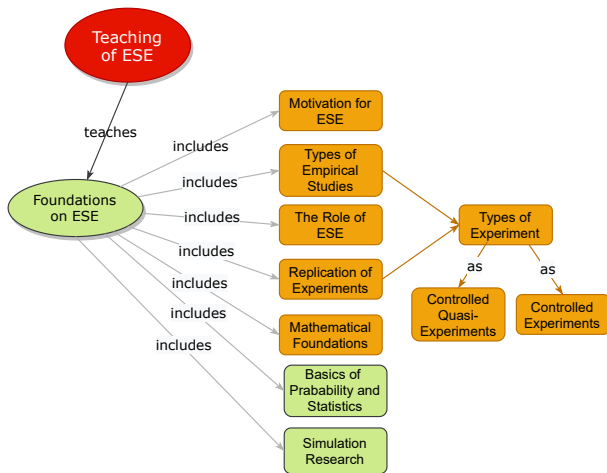


Figure 3: Relationships of the Concept Foundations on ESE

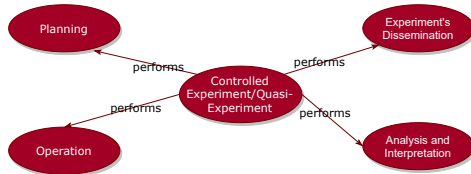


Figure 4: (Quasi-)Experiment Concepts Grouping. Adapted from Furtado et al. [9]

the learning results. According to Prentice and Robinson [21] there is ongoing pressure to identify possible learning materials. Thus, the need for instructors to develop or adopt additional methods to document students learning objectively is a challenge. Most of these approaches have focused on measuring student learning at the specific course level.

Figure 5 depicts the relationships among possible methods of evaluation of learning outcomes. Such methods are explained as follows:

- **Experiment Carry Out:** means a practical conduction of a controlled experiment by students;
- **Seminar:** refers to the academic presentation of research results or even ESE teaching contents, in which students debate on the topic;
- **Paper:** means the production of a scientific article, based on bibliographic research, experience report in conducting experiments, or the complete writing of the practical application of experimentation;
- **Technical Report:** is the writing of reports or abstracts, involving books, articles, or chapters on concepts and fundamentals of experimentation.

### 3.4 Teaching Material Concept

To provide quality education to students, the use of teaching materials becomes a fundamental part of the ESE teaching-learning process. The teaching materials should not be used in an ad-hoc

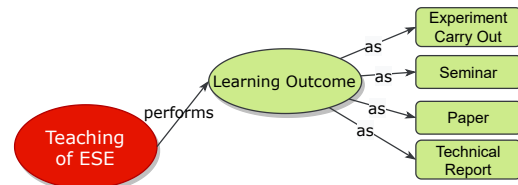


Figure 5: Relationships of the Concept Learning Outcome

way. There must be planning on the part of the instructor, who must know how to use them to achieve the objective proposed by each ESE course. The material to be used must provide the student with a stimulus to research and search for new knowledge, the purpose of using concrete materials is to make the students acquire an investigative culture which will prepare them to face the world with practices as an active subject in society [19]. Therefore, we derived a list of possible Teaching Material to be used by the instructors, depicted in Figure 6, and explained as follows:

- **Reference Text:** refers to classical texts from Book, Book Chapter, or Paper, which are widely known and referred to in the SE experimentation area. For instance, the book by Wohlin et al. [29]<sup>2</sup>; and
- **Other Material:** refers to different types of materials and resources that can be used by instructors to support the teaching of ESE. This concept could be represented as a Book, a Book Chapter, a Paper, Class Notes, Invited Talks, or Slides.

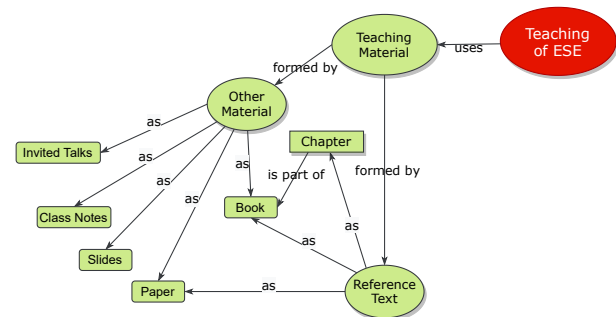


Figure 6: The Teaching Material Concept

### 3.5 Open Educational Resource (OER) Concept

As important as the use of teaching materials by instructors in the teaching-learning process, is how such materials can be found, consumed, and manipulated. One of the forms of open distribution of teaching materials is through the classical concept of Open Educational Resources (OERs).

According to Hylén [13] there are many critical issues around access, quality, and cost of information and knowledge on the Internet, as well as the provision of content and teaching material. As it becomes clearer that the growth of the Internet offers real

<sup>2</sup>In Luz et al. [19] we provide a list of all books surveyed.

opportunities to improve access and transfer of knowledge and information from universities and colleges to a wider range of users, there is an urgent need to clarify these issues with a special focus on initiatives of Open Educational Resources.

Figure 7 shows the organization of materials based on OERs. We defined such concepts based on studies carried out in the literature, with the aim of providing instructors with a basic structure containing aspects about a possible organization of the distribution of teaching materials, which can be used to support ESE teaching.

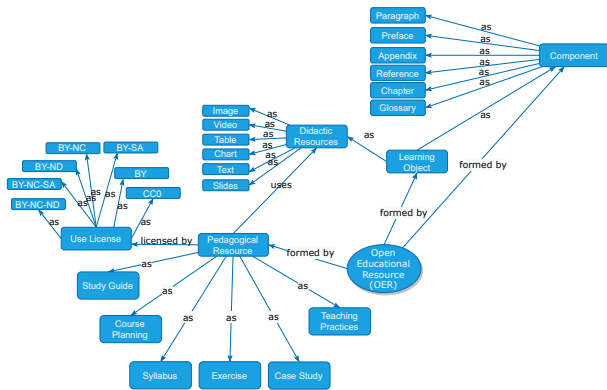


Figure 7: OER-based Teaching Material Organization

Open Educational Resource (OER) could be formed by (one or more):

- **Component:** can be represented as a Paragraph, a Preface, an Appendix, a Reference, a Chapter, or a Glossary;
- **Learning Object:** is a pedagogical resource that can be used by instructors and represented as a Component or a Didactic Resource.
- **Pedagogical Resource:** are teaching resources in the form of a Study Guide, Course Planning, Syllabus, Exercise, Case Study, or Teaching Practices. Pedagogical resources should be attributed a Use License, which might be of any of the following types: CC0, BY, BY-SA, BY-NC, BY-ND, BY-NC-SA, or BY-NC-ND.

The Didactic Resources concept uses Pedagogical Resource and can also be represented as Image, Video, Table, Chart, Text, or Slides.

## 4 EMPIRICAL EVALUATION

This section presents a CM empirical study based on the Technology Acceptance Model (TAM) [6].

### 4.1 Aim

This study is **aimed** at analyzing the specified CM **with the purpose of** characterizing it **with relation to** the TAM dimensions of Perceived Ease of Use, Perceived Usefulness, Attitude Toward Using, and Actual System Use **from the point of view of** ESE researchers **in the context of** instructors who teach ESE concepts.

Such dimensions are as follows [6]:

- **Perceived Ease of Use (EoU)** is the degree to which a person believes that using a particular system would be trouble-free;

- **Perceived Usefulness (Usef)** is the degree to which each person believes that using a particular system could increase their professional performance;
- **Attitude Toward Using (AtU)** is the individual degree of evaluation that will influence the behavior intention. This dimension is related to how the CM can influence its use by instructors;
- **Actual System Use (AsU)** is the use of the proposed CM in the studied domain.

We defined 19 multiple-choice questions regarding all four dimensions. Each question could be answered on the following Likert scale: totally agree, agree, neutral, disagree, and totally disagree. Table 1 presents such questions.

We also defined seven open-ended questions (OQ.1 through OQ.8), thus the participants might express their opinions on the CM, as follows:

- **OQ.1** What is your general opinion about the PERCEIVED EASE OF USE of the CM?
- **OQ.2** What is your general opinion about the PERCEIVED USEFULNESS of the CM?
- **OQ.3** What is your general opinion about the ATTITUDE TOWARD USING the CM?
- **OQ.4** What is your general opinion about the ACTUAL SYSTEM USE of the CM?
- **OQ.5** What are the positive points of the CM to support the teaching of ESE?
- **OQ.6** What improvement points do you highlight to the CM conceptual model to support the teaching of ESE?
- **OQ.7** What issues/problems/concerns do you highlight to the CM to support the teaching of ESE?

### 4.2 Planning

For the planning of this study, we defined training, selection of participants, and instrumentation. To support **training**, we recorded videos<sup>3</sup> and made them openly available to guide participants on this study phases, as follows:

- **Characterization and Rationale of the CM**, including research problem and the CM explanation;
- **The CM Design** with the steps taken in the design of the CM;
- **Description of the CM and Relationships** containing explanations on the relationships and groups of concepts in the CM;
- **Examples of Instances of the CM** with several examples of instances to be performed by participants in the proposed CM;
- **Steps to Carry out this CM Empirical Study** with tasks to be carried out by the participants.

In the **selection of participants**, an analysis of potential participants was carried out by accessing the websites of the graduate programs in which they were enrolled. The list of Brazilian graduate programs was obtained from the Sucupira platform, the Brazilian platform for graduate programs' quality evaluation. The participant contact emails were obtained through the websites of the graduate

<sup>3</sup>In Portuguese with English subtitles by Youtube.



**Table 1: Questions for the TAM Dimensions**

| TAM Dimension                | Questions                                                                                                                                                 |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Perceived Ease of Use</b> | <b>EoU.1</b> I find the CM easy to use.                                                                                                                   |
|                              | <b>EoU.2</b> Learning to use/interpret the CM was easy.                                                                                                   |
|                              | <b>EoU.3</b> The use of the CM facilitates the structuring/proposing ESE courses.                                                                         |
|                              | <b>EoU.4</b> I find the CM easy to understand.                                                                                                            |
|                              | <b>EoU.5</b> It took a long time for me to understand and instantiate the CM.                                                                             |
| <b>Perceived Usefulness</b>  | <b>Usef.1</b> Using the CM might aid me to develop OER for teaching ESE.                                                                                  |
|                              | <b>Usef.2</b> Using the CM might contribute to the quality of my work.                                                                                    |
|                              | <b>Usef.3</b> Using the CM will be important for the execution of my teaching activities in ESE.                                                          |
|                              | <b>Usef.4</b> The CM will be useful for my ESE teaching activities.                                                                                       |
|                              | <b>Usef.5</b> The proposed CM will add value to my teaching work on ESE.                                                                                  |
| <b>Attitude Toward Using</b> | <b>AtU.1</b> I consider the CM and its elements positive for teaching ESE.                                                                                |
|                              | <b>AtU.2</b> The CM is compatible with other teaching methods I use in my daily life.                                                                     |
|                              | <b>AtU.3</b> I can easily aggregate or compare the data in the CM with the materials I currently use.                                                     |
|                              | <b>AtU.4</b> I believe it will be interesting to use the CM as a support in my teaching activities in ESE.                                                |
|                              | <b>AtU.5</b> I want to use the CM for my tasks in addition to current methods of teaching ESE.                                                            |
| <b>Actual System Use</b>     | <b>AsU.1</b> The CM can be applied to represent concepts for teaching ESE.                                                                                |
|                              | <b>AsU.2</b> The CM facilitates the understanding of vocabularies, in addition to the standardization of concepts to be worked on in the teaching of ESE. |
|                              | <b>AsU.3</b> The use of the CM allows for organizing and complementing my course with additional content for teaching ESE.                                |
|                              | <b>AsU.4</b> The CM can support as a methodological basis the elaboration of experimentation courses in Software Engineering.                             |

programs, personal websites, or published papers. The criterion used for the selection of participants was whether they teach or had taught ESE in graduate programs. We then selected 104 potential participants. After a careful analysis of their research profiles, we came up with 78 participants to invite. We invited them via email. After sending the invitation, 13 agreed to take part in the evaluation, but only nine did it.

The participants have a mean value of 11.77 years of experience in software engineering, with a median of 12 years, standard deviation of 2.7 years, max of 18 years, and min of 8 years. Regarding the experience in ESE, they have a mean value of 5 years, median of 4 years, standard deviation of 2.95 years, max of 12 years, and min of 2 years. In addition, two participants are female and seven are male. Regarding Brazilian states, one from DF, five from PR, one from RS, one from SC, and one from SP. We will include this information in the camera-ready version.

All participants had electronic access to the following **instruments**, in Portuguese:

- Free Informed Consent Term;
- Instructional videos on research and training on how to proceed with the CM evaluation;
- Slides used in the videos;
- A file containing the proposed CM for evaluation;
- A document with the glossary of the concepts contained in the CM;
- A worksheet for the CM instantiation; and
- An online questionnaire with TAM dimension-related questions.

### 4.3 Operation

Firstly, we carried out a pilot with an instructor from the Federal University of Technology, Brazil. The data obtained were discarded, however, his/her considerations indicated improvements, which we implemented. The subsequent steps of the execution of this study occur as follows:

- (1) the participants accessed the links sent via email to the study instrumentation;
- (2) the participants read the necessary documents;
- (3) the participants accessed the document containing the CM;
- (4) the participants followed each step to instantiate their ESE courses using the CM;
- (5) the participants answered the questionnaire with TAM-related questions;
- (6) the participants sent the answers via Google Forms.

The participants had 15 days to complete the online evaluation questionnaire. All the study follow-up was conducted online because of the COVID-19 pandemic. Basically, it was not necessary to validate the data provided by the participants. The data from the open questions were all validated by carefully reading the answers. In the end, all the data could be used in this evaluation.

### 4.4 Analysis and Interpretation

We analyzed the Likert-based questions with percentages and modes.

**4.4.1 General Analysis of the TAM Dimensions.** Table 2 presents the analysis for each TAM dimension.

The agreement for **Perceived Ease of Use** was 64.5% (37.8% + 26.7%). We can observe none participants totally disagree with this

**Table 2: TAM-based answers summary**

| Dimension             | TD   | D     | N     | A     | TA    |
|-----------------------|------|-------|-------|-------|-------|
| Perceived Ease of Use | 0.0% | 15.5% | 20.0% | 37.8% | 26.7% |
| Perceived Usefulness  | 0.0% | 4.4%  | 22.2% | 57.8% | 15.6% |
| Attitude Toward Using | 2.2% | 4.4%  | 15.6% | 48.9% | 28.9% |
| Actual System Use     | 0.0% | 2.7%  | 11.2% | 36.1% | 50.0% |

*TD = totally disagree, D = disagree, N = neutral, A = agree, TA = totally agree*

dimension, only 15.5% disagree, and 20% were neutral, thus totaling 35.5%, one-third of the participants.

When analyzing the **Perceived Usefulness** dimension, 73.4% (57.8% + 15.6%) of participants agree with it, while none totally disagree. In addition, 4.4% disagree and 22.2% are neutral, which corresponds to 26.6%, practically one-fourth of the sample.

The **Attitude Toward Using** agreement was 77.8% (48.9% + 28.9%) of the sample, while 2.2% totally disagree and 4.4% disagree. 15.6% were neutral, totaling 22.2%, almost one-fourth of the sample.

Analyzing the **Actual System Use** dimension, we can see that 86.1% (36.1% + 50%) agree with it, whereas none totally disagree, 2.7% disagree, and 11.2% are neutral, which corresponds to 23.9% less than one-fourth of the sample.

Generally analyzing, the CM had an interesting acceptance in all four TAM dimensions. A positive point to consider in this general analysis is that the dimensions have at least 64.5% of agreement, which is important evidence of the CM feasibility.

**4.4.2 Individual Analysis of the TAM Dimensions.** Table 3 presents the answers for each TAM dimension per participant and respective mode measure.

**Table 3: TAM Dimension Answers per Participant (P.1 through P.9)**

| Question | P1 | P2 | P3 | P4 | P5 | P6 | P7 | P8 | P9 | MODE  |
|----------|----|----|----|----|----|----|----|----|----|-------|
| EoU.1    | 5  | 4  | 3  | 4  | 3  | 4  | 5  | 5  | 5  | 5     |
| EoU.2    | 5  | 4  | 2  | 4  | 3  | 2  | 5  | 4  | 5  | 4/5   |
| EoU.3    | 3  | 4  | 2  | 3  | 4  | 5  | 5  | 4  | 5  | 4/5   |
| EoU.4    | 4  | 4  | 4  | 4  | 4  | 2  | 3  | 5  | 5  | 4     |
| EoU.5    | 2  | 2  | 3  | 4  | 2  | 4  | 3  | 3  | 4  | 2/3/4 |
| Usef.1   | 3  | 4  | 4  | 3  | 3  | 4  | 5  | 4  | 4  | 4     |
| Usef.2   | 3  | 4  | 3  | 4  | 5  | 5  | 4  | 5  | 5  | 5     |
| Usef.3   | 4  | 4  | 4  | 3  | 4  | 4  | 2  | 4  | 4  | 4     |
| Usef.4   | 4  | 4  | 3  | 4  | 4  | 5  | 2  | 5  | 3  | 4     |
| Usef.5   | 3  | 4  | 3  | 4  | 4  | 4  | 4  | 4  | 4  | 4     |
| AtU.1    | 3  | 4  | 4  | 5  | 4  | 4  | 4  | 5  | 5  | 4     |
| AtU.2    | 5  | 2  | 4  | 4  | 3  | 3  | 5  | 4  | 4  | 4/5   |
| AtU.3    | 3  | 4  | 4  | 4  | 4  | 2  | 4  | 5  | 5  | 4     |
| AtU.4    | 3  | 4  | 4  | 4  | 4  | 4  | 4  | 5  | 5  | 4     |
| AtU.5    | 1  | 4  | 3  | 5  | 3  | 4  | 4  | 5  | 5  | 4/5   |
| AsU.1    | 5  | 4  | 3  | 5  | 5  | 5  | 5  | 5  | 5  | 5     |
| AsU.2    | 5  | 4  | 4  | 4  | 5  | 4  | 5  | 4  | 5  | 4     |
| AsU.3    | 3  | 4  | 4  | 5  | 3  | 4  | 2  | 5  | 5  | 4/5   |
| AsU.4    | 4  | 4  | 3  | 5  | 5  | 4  | 5  | 4  | 5  | 4/5   |

*Totally Disagree = 1, Disagree = 2, Neutral = 3, Agree = 4, Totally Agree = 5*

When analyzing Table 4 for the **Perceived Ease of Use** dimension, we observe that **EoU.1** had 77.7% agreement as the participants

find the CM ease to use. For **EoU.2** and **EoU.3**, 66.6% of them understand learning to use and interpret the CM is easy and it facilitates the structuring/proposing of ESE courses, respectively. Regarding **EoU.4** 77.8% agree the CM is easy to understand, whereas only 33.3% find they will take a long time to understand and instantiate such CM **EoU.5**. For this dimension specifically, disagreement among participants had low rates, which corroborates our general findings from Section 4.4.1. Note that no answers were given for the option “totally disagree” in this dimension.

**Table 4: Answers per TAM Dimension**

| Question                     | TD    | D     | N     | A     | TA    |
|------------------------------|-------|-------|-------|-------|-------|
| <b>Perceived Ease of Use</b> |       |       |       |       |       |
| EoU.1                        | 0.0%  | 0.0%  | 22.2% | 33.3% | 44.4% |
| EoU.2                        | 0.0%  | 22.2% | 11.1% | 33.3% | 33.3% |
| EoU.3                        | 0.0%  | 11.1% | 22.2% | 33.3% | 33.3% |
| EoU.4                        | 0.0%  | 11.1% | 11.1% | 55.6% | 22.2% |
| EoU.5                        | 0.0%  | 33.3% | 33.3% | 33.3% | 0.0%  |
| <b>Perceived Usefulness</b>  |       |       |       |       |       |
| Usef.1                       | 0.0%  | 0.0%  | 33.3% | 55.6% | 11.1% |
| Usef.2                       | 0.0%  | 0.0%  | 22.2% | 33.3% | 44.4% |
| Usef.3                       | 0.0%  | 11.1% | 11.1% | 77.8% | 0.0%  |
| Usef.4                       | 0.0%  | 11.1% | 22.2% | 44.4% | 22.2% |
| Usef.5                       | 0.0%  | 0.0%  | 22.2% | 77.8% | 0.0%  |
| <b>Attitude Toward Using</b> |       |       |       |       |       |
| AtU.1                        | 0.0%  | 0.0%  | 11.1% | 55.6% | 33.3% |
| AtU.2                        | 0.0%  | 11.1% | 22.2% | 33.3% | 33.3% |
| AtU.3                        | 0.0%  | 11.1% | 11.1% | 55.6% | 22.2% |
| AtU.4                        | 0.0%  | 0.0%  | 11.1% | 66.7% | 22.2% |
| AtU.5                        | 11.1% | 0.0%  | 22.2% | 33.3% | 33.3% |
| <b>Actual System Use</b>     |       |       |       |       |       |
| AsU.1                        | 0.0%  | 0.0%  | 11.1% | 11.1% | 77.8% |
| AsU.2                        | 0.0%  | 0.0%  | 0.0%  | 55.6% | 44.4% |
| AsU.3                        | 0.0%  | 11.1% | 22.2% | 33.3% | 33.3% |
| AsU.4                        | 0.0%  | 0.0%  | 11.1% | 44.4% | 44.4% |

*TD = totally disagree, D = disagree, N = neutral, A = agree, TA = totally agree*

For the **Perceived Usefulness** dimension, 66.7% (55.6% + 11.1%) of participants agree that the use of the CM might aid them to develop OER for teaching ESE (**Usef.1**), whereas 33.3% are indifferent to this matter. Related to the contribution of using the CM for improving the quality of the participants’ work **Usef.2**, 77.7% (33.3% + 44.4%) agree with it, whereas 22.2% are indifferent. In **Usef.3** most of the participants (77.8%) agree with using the CM will be important for the execution of their teaching activities. For the same question, on the other hand, only 11.1% disagree. Regarding **Usef.4**, 66.6% agree the CM will be useful for their ESE teaching activities, while only 11.1% disagree. Considering that the CM will add value to the participants’ work on ESE **Usef.5**, 77.8% agree with it, whereas none disagree. Note that no answers were given for the option “totally disagree” in this dimension, as occurred for “Perceived Ease of Use”.

Table 4 presents answers to the **Attitude Toward Using** dimension. Most of the participants (88.9%) consider the CM and its elements positive for teaching ESE and none of them disagrees with it (**AtU.1**). Regarding the CM being compatible with other teaching methods that participants use in their daily life (**AtU.1**), 66.6% agree, whereas 22.2% were indifferent or disagree (11.1%). In **AtU.3**, 77.8% agree they can easily aggregate or compare the data in the CM with materials they currently use, while only 11.1% disagree or is indifferent (11.1%). For **AtU.4**, the majority of the



participants (88.9%) believe it will be interesting to use the CM as a support in their teaching activities in ESE, whereas 11.1% are indifferent. Regarding the participants' willingness to use the CM for their tasks in addition to current methods of teaching ESE (**AtU.5**), 66.6% agree, whereas 22.2% is indifferent or disagree (11.1%). Note that, for this dimension, only one question (**AtU.5**) had a "totally disagree" answer, and only two questions (**AtU.2** and **AtU.3**) had "disagree" answers.

Table 4 presents the **Actual System Use** dimension answers. In **AsU.1** we asked whether the CM can be applied to represent concepts for teaching ESE. Thus, 88.9% agree with it, whereas 11.1% are indifferent. Regarding **AsU.2**, all participants answered the CM facilitates the understanding of vocabularies, in addition to the standardization of concepts to be worked on in the teaching of ESE. As a matter of the CM used to allow organizing and complementing the participants' courses with additional content for teaching ESE **AsU.3**, most (66.6%) of them agree, whereas 22.2% are indifferent or disagree (11.1%). Regarding the CM can support as a methodological basis the elaboration of experimentation courses in Software Engineering (**AsU.4**), 88.8% of the participants agree, while 11.1% are indifferent. For this dimension, note no answers were given to "totally disagree" and only to "disagree".

**4.4.3 Open-ended Answers Analysis.** We analyzed the open-ended questions by coding the quotes of each participant, thus highlighting the most important ones.

With the analysis of **OQ.1**, we observed that, in general, the participants' opinions were encouraging, such as: *"The conceptual model is relatively easy to understand"*, *"The structure and precision of the CM facilitate its use"*, and *"The model is relatively easy to use, although it involves many concepts, all of which are pertinent in my opinion. Its overall, usage is easy"*.

The **OQ.2** analysis gave us a general understanding that the participants' opinions are favorable to the CM perceived usefulness, exemplified by: *"It [the CM] is concise, allows for a relatively modular presentation, and would facilitate communication with students"*; *"Certainly, the template is useful. In the case of my course, it covers a good introductory part, in addition to the specific part of controlled experiments and quantitative analysis"*; and *"I find the idea very interesting and it certainly helps to guide the planning of content and teaching resources"*.

Question **OQ.3** presents the participants' view of the Attitude Toward Using of the CM. We had interesting responses, such as: *"The main reason [of the CM] is the easiness at presenting its concepts"* and *"when performing the evaluation [of the CM] I was able to anticipate things I can do to improve my course, including related to other types of empirical studies"*.

When we analyze **OQ.4** on the CM actual system use, we observed an interesting point: *"I believe that the graphic representation has an important didactic character for present the concepts, as well as their relationship with the pedagogical part"*.

Question **OQ.5** deals with the strengths of the CM. It is clear for us the CM is seen as a resource that will facilitate the exposure of ESE concepts to students. The CM was also understood by the participants as a quick reference guide on experiments. It was reported that the CM contemplates all the concepts of experimentation, as well as its process. The CM is also important as a

reference for the creation of courses, and elaboration of teaching plans, methodologies, and lesson plans for ESE.

Regarding the improvement points (**OQ.6**) for the CM, we observed the participants pointed out the need for the CM to present a complementary procedural or at least ordered view, thus the user can know what to do and in what order. Another improvement presented would be the refinement of certain relationships, indicating the traceability among elements. Certain participants highlighted the representation used by the CM in the form of ellipses and rectangles, seeking to enumerate each step of the process, with the purpose of facilitating the teaching and understanding of the students. A given participant indicated the evolution of the CM to an ontology.

When reporting the negative points observed in **OQ.7**, there was only one point to be discussed: the form of presentation of the CM. Such a way could consider a top-down approach, to mitigate fatigue by presenting the same. Therefore, this point should be addressed in future works.

## 4.5 Discussion of Results

Based on the participants' answers on the CM perceived ease of use, participants considered the CM easy to understand and use, providing no serious difficulties in learning to use it or interpret its concepts. In addition, it is able to be used as an instrument to facilitate the structuring of new ESE courses. Overall, the CM's ease of use received positive and encouraging feedback.

Regarding the perceived usefulness dimension, in certain tasks that involve the preparation of materials or the CM use as an aid to participants in their activities, few participants disagree. Therefore, we used these concerns as a guide to evolve the prospectively CM. Even so, it is clear that most participants found the CM useful.

As a matter of the attitude toward using the CM, we understand the CM could really be considered interesting enough to further be used by the participants. This is corroborated by several answers as the participants found the CM elements positive for the teaching of ESE, whereas they look to incorporating the CM in their academic activities and using it as a support resource in ESE classes. However, important points were raised by the participants, such as, the CM being seen as a tool to guide the planning of a course and not directly assisting in the teaching-learning process. Although the purpose of the CM is to be a support resource, certain pedagogical elements that might affect the teaching-learning process were not considered so far.

The perception of CM actual system use is very optimistic, as we obtained an interesting acceptance of it, especially because the CM aids instructors in guiding them in defining concepts to be taught, an easy understanding of vocabularies, and standardization of concepts to be worked. In addition, it serves as a methodological basis for the elaboration of new ESE courses.

In summary, the use of a questionnaire with multiple choice and open-ended questions brought the opportunity to have a concise view of the CM. At the same time, it allowed participants to freely express their opinions, thus identifying issues, inconsistencies, positives, negatives, and improvement points for our CM.

## 4.6 Threats to Validity

**Construct Validity:** the questionnaire was built based on aspects related to the clarity and objectivity of the CM instantiated as an element of support and consultation to the teaching of ESE. We might have missed certain fundamental aspects of such an element.

**Internal validity:** the accuracy of the participants' answers is considered valid since the conceptual model was evaluated by ESE instructors. The effect of fatigue must be taken into account, as the evaluation of the CM consists of videos and supporting materials. Thus, to minimize this threat, the explanations about the CM and other actions to be carried out in the evaluation were recorded and made available in five short videos of up to 14 minutes. In addition, the documentation was sent to the participants by email with an interval of 15 days to return the answers.

**External validity:** there was considerable difficulty in finding instructors who work or have worked in ESE, amongst the 82 higher education institutions in Brazil, listed on the Sucupira<sup>4</sup>. However, we understand our sample is very accurate in their responses, thus contributing to improving our CM.

## 5 CONCLUSION

We presented a conceptual model to support the teaching of Controlled Experimentation in Software Engineering. The CM was created after an analysis of different sources, including a survey on teaching ESE.

We also evaluated the CM regarding four dimensions based on the TAM model with researchers who teach ESE. We could identify several points in favor of our CM. However, actionable issues to improve such CM were also identified, thus we took them into consideration to evolve it.

As future work we intend to first evolve our CM with the results discussed in this study, thus empirically evaluating it again inviting other instructors and start investigating an ontology to build a (semi-)automatized supporting tool for instructors to benefit from our model.

## ARTIFACT AVAILABILITY

All CM images are available in high-resolution at <https://doi.org/10.5281/zenodo.7692078>.

Questions and data from the empirical study are available at <https://doi.org/10.5281/zenodo.7991913>.

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<sup>4</sup><https://sucupira.capes.gov.br/sucupiraplatform>